

## **Industrial Internship Report on "Multi-stage continuous-flow manufacturing process - Real process data to predict factory output"**

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### *Executive Summary*

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Multi-stage continuous-flow manufacturing process - Real process data to predict factory output." The project uses process readings from a continuous manufacturing line to predict its output measurements. I explored the dataset and tested Linear Regression and Random Forest models for selected measurements.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

## TABLE OF CONTENTS

|     |                                              |                                     |
|-----|----------------------------------------------|-------------------------------------|
| 1   | Preface .....                                | 3                                   |
| 2   | Introduction .....                           | 4                                   |
| 2.1 | About UniConverge Technologies Pvt Ltd ..... | 4                                   |
| 2.2 | About upskill Campus .....                   | 8                                   |
| 2.3 | Objective .....                              | 10                                  |
| 2.4 | Reference.....                               | 10                                  |
| 2.5 | Glossary .....                               | 10                                  |
| 3   | Problem Statement .....                      | 11                                  |
| 4   | Existing and Proposed solution .....         | 12                                  |
| 5   | Proposed Design/ Model .....                 | 13                                  |
| 5.1 | High Level Diagram (if applicable) .....     | 13                                  |
| 5.2 | Low Level Diagram (if applicable) .....      | <b>Error! Bookmark not defined.</b> |
| 5.3 | Interfaces (if applicable) .....             | 13                                  |
| 6   | Performance Test .....                       | 14                                  |
| 6.1 | Test Plan/ Test Cases .....                  | 14                                  |
| 6.2 | Test Procedure.....                          | 15                                  |
| 6.3 | Performance Outcome .....                    | 15                                  |
| 7   | My learnings .....                           | 17                                  |
| 8   | Future work scope .....                      | 18                                  |

## 1 Preface

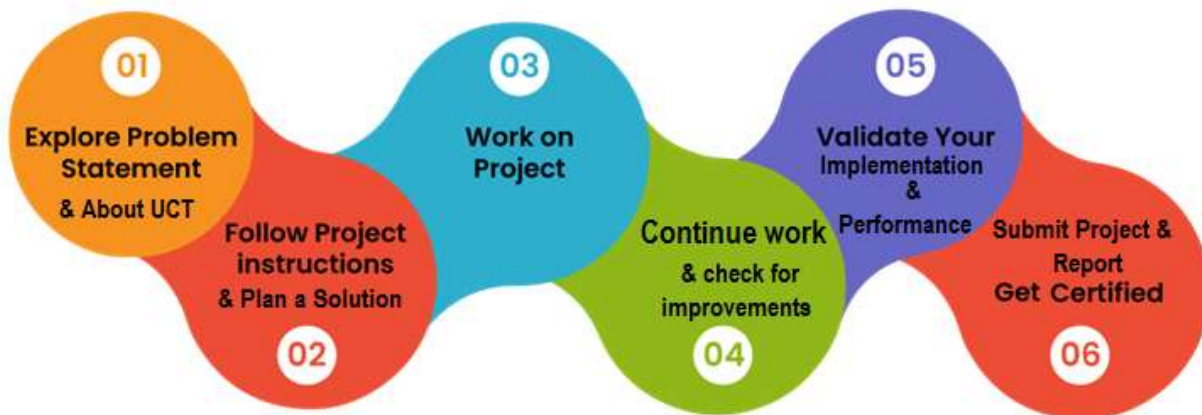
This report covers my work on a multi-stage factory output prediction project, including data exploration, cleaning, baseline modeling, lag features, selected Stage 2 tests, and a basic anomaly-flagging experiment. The earlier weekly reports document the stages of this work.

About need of relevant Internship in career development.

The project studies factory readings to predict output measurements. Machines 1, 2 and 3 feed a combiner in the first stage; Machines 4 and 5 then process the material in series. The 15 measurements after the combiner are the main targets.

Opportunity given by USC/UCT.

How Program was planned



I improved my practical use of pandas, NumPy and scikit-learn for cleaning data, building models and checking errors. The experience showed me why industrial data must be checked carefully before its results can be trusted.

Thank to all (with names), who have helped you directly or indirectly.

Your message to your juniors and peers.

## 2 Introduction

### 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT)**, **Cyber Security**, **Cloud computing (AWS, Azure)**, **Machine Learning**, **Communication Technologies (4G/5G/LoRaWAN)**, **Java Full Stack**, **Python**, **Front end** etc.



#### i. UCT IoT Platform ()

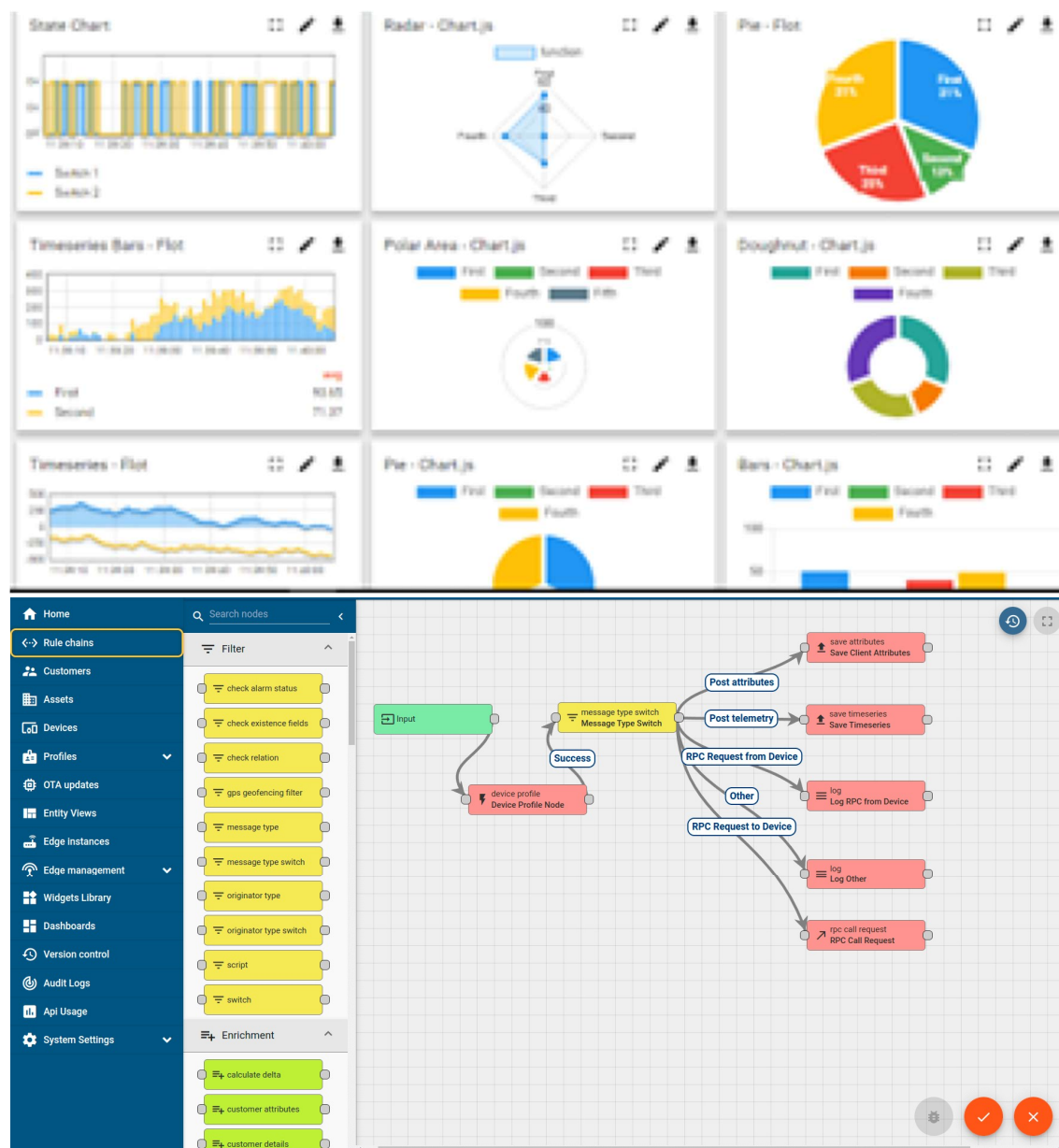
**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



## FACTORY WATCH

### ii. Smart Factory Platform ( )

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleashed the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



| Machine   | Operator   | Work Order ID | Job ID | Job Performance | Job Progress |          | Output  |        | Rejection | Time (mins) |      |          |      | Job Status  | End Customer |
|-----------|------------|---------------|--------|-----------------|--------------|----------|---------|--------|-----------|-------------|------|----------|------|-------------|--------------|
|           |            |               |        |                 | Start Time   | End Time | Planned | Actual |           | Setup       | Pred | Downtime | Idle |             |              |
| CNC_S7_81 | Operator 1 | WO0405200001  | 4168   | 58%             | 10:30 AM     |          | 55      | 41     | 0         | 80          | 215  | 0        | 45   | In Progress | i            |
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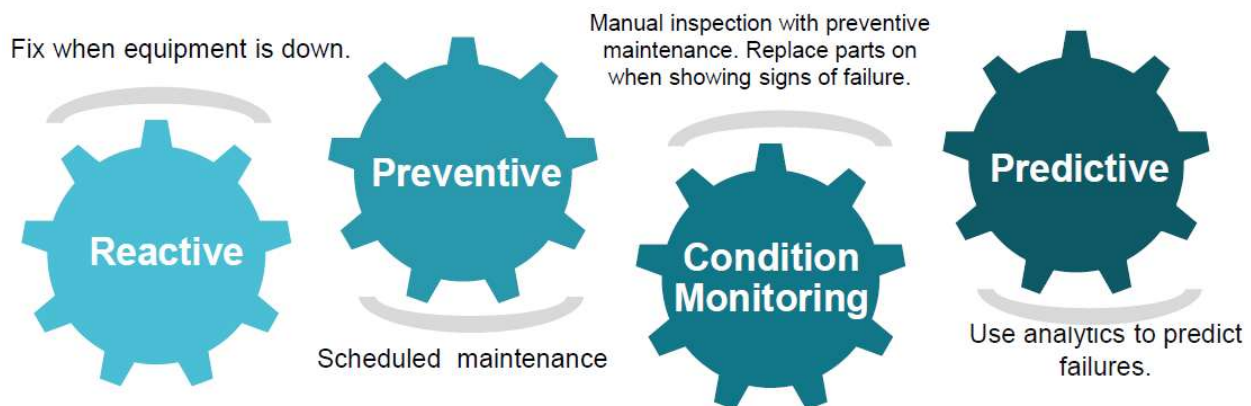


### iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

### iv. Predictive Maintenance

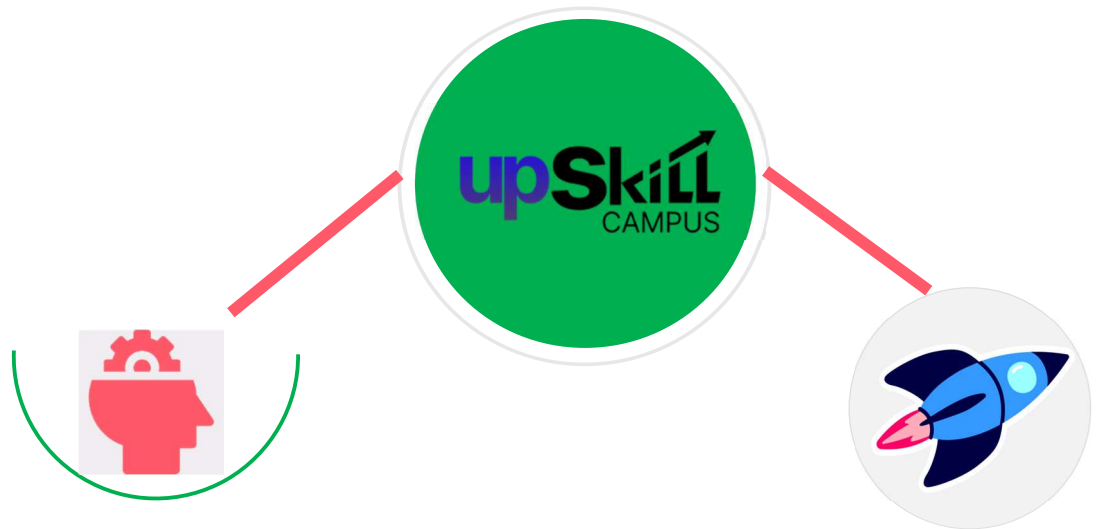
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



## 2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

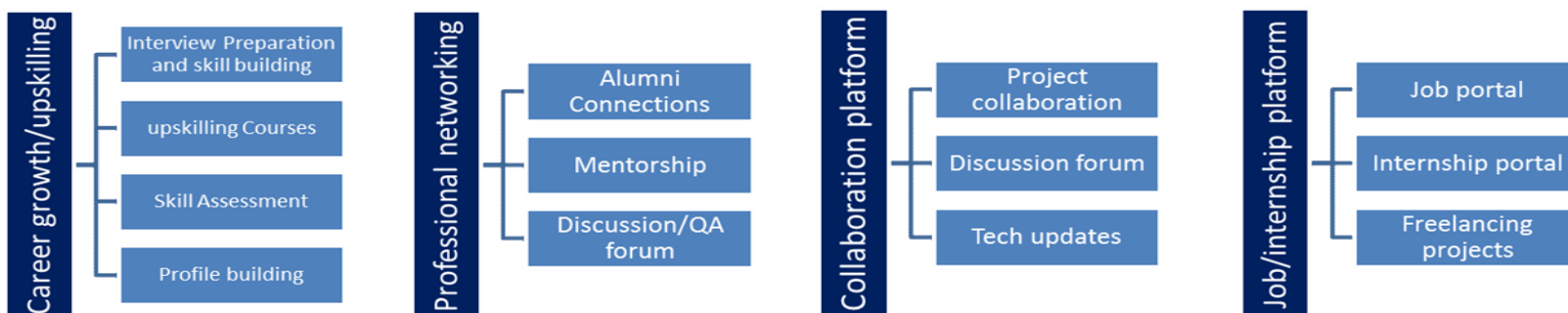
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



## 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

## 2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

## 2.5 Reference

- [1] Project dataset: continuous\_factory\_process.csv
- [2] Dataset notes: notes\_on\_dataset.txt
- [3] scikit-learn official documentation
- [4] UCT and upskill Campus internship material

## 2.6 Glossary

| Terms    | Acronym                                                       |
|----------|---------------------------------------------------------------|
| CSV      | Comma-separated values file                                   |
| MAE      | Mean absolute error                                           |
| $R^2$    | Score measuring explained variation                           |
| Time lag | Earlier machine readings used to predict a later output       |
| Anomaly  | An unusual reading that differs greatly from model prediction |

### 3 Problem Statement

In the assigned problem statement

The manufacturing line is a high-speed continuous process with multiple machines and stages. The main problem is to predict important factory output measurements using machine process data.

In the first stage, Machines 1, 2, and 3 operate in parallel. Their outputs are combined in a combiner stage. After the combiner, the output is measured at 15 locations. These Stage 1 measurements are the primary outputs to predict.

The material then enters the second stage, where Machines 4 and 5 operate in series. The output is again measured at 15 locations. These Stage 2 measurements are secondary outputs and are more difficult to predict because the data is noisier.

The challenge is to use machine temperature, pressure, motor speed, material properties, combiner settings, and earlier time readings to predict the factory output. A successful model can support real-time quality monitoring, process control, and anomaly detection.

## 4 Existing and Proposed solution

A simple baseline is Linear Regression using machine-process variables. This is easy to understand, but a straight-line model may miss more complex patterns and time delays. The dataset does not document an existing factory deployment, so no claim is made about replacing a live system.

I loaded the CSV with pandas, checked the column groups and timestamps, examined invalid negative readings, and built a Linear Regression baseline. I then tested Random Forest with current and 1–2-row-old machine readings on selected Stage 1 targets. Selected Stage 2 targets were explored using Stage 1 outputs and Machines 4–5 readings as inputs.

The main value is an initial comparison of modeling approaches and a way to identify readings needing review. These are offline experiments, not a verified real-time control system.

### 4.1 Code submission (GitHub link):

<https://github.com/bhageerath78236/upskillcampus/blob/main/MultiStageContinuousFlowManufacturingProcess.ipynb>

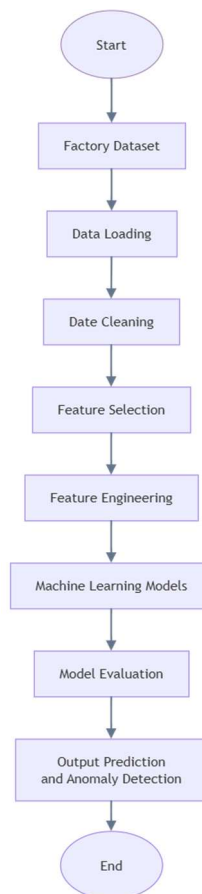
### 4.2 Report submission (GitHub link):

[https://github.com/bhageerath78236/upskillcampus/blob/main/MultiStageContinuousFlowManufacturingProcess\\_Bhageerath\\_USC\\_UCT.pdf](https://github.com/bhageerath78236/upskillcampus/blob/main/MultiStageContinuousFlowManufacturingProcess_Bhageerath_USC_UCT.pdf)

## 5 Proposed Design/ Model

CSV file → check data types and output columns → parse timestamps → inspect or exclude questionable output readings → choose upstream machine variables → add earlier-row features → train Linear Regression and Random Forest → compare MAE and  $R^2$  → inspect unusually large prediction errors. Only selected targets were tested with Random Forest.

### 5.1 High Level Diagram



**Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM**

### 5.2 Interfaces

Notebook input: continuous\_factory\_process.csv. Notebook output: summary tables, prediction scores and experimental anomaly flags. No live sensor interface, protocol integration or production dashboard was implemented.

## 6 Performance Test

Offline tests were used to check prediction error and compare models. A live-system test was not performed.

Constraints include noisy measurements, possible invalid sensor readings, data timing and the runtime needed to train models.

Negative Stage 1 output values were excluded for the reported modeling tests. Earlier-row features were tested as a first step toward representing delay, although row shifts are not guaranteed to equal exact seconds.

On a selected 4,998-row working sample, Random Forest obtained  $R^2$  scores of 0.867, 0.872 and 0.895 for Stage 1 Measurements 6, 11 and 14. These were offline random-split results and are not proof of future or live performance.

Runtime and memory in a live production environment were not measured. Random Forest training on larger sets took longer in the earlier experiments.

Before deployment, use a time-ordered holdout, verify sensor values and actual sampling times, benchmark prediction latency, and test alerts against confirmed faults.

### 6.1 Test Plan/ Test Cases

| Test Case            | Input                      | Expected Result                               |
|----------------------|----------------------------|-----------------------------------------------|
| Dataset loading      | CSV file                   | Dataset loads successfully                    |
| Missing-value check  | Full dataset               | Missing values are identified if present      |
| Negative-value check | Stage 1 outputs            | Invalid negative values are found and cleaned |
| Timestamp conversion | Timestamp column           | Timestamp converts to datetime format         |
| Baseline model       | Stage 1 inputs and outputs | Linear Regression scores are generated        |

| Test Case      | Input                                     | Expected Result                                  |
|----------------|-------------------------------------------|--------------------------------------------------|
| Improved model | Lag features and Random Forest            | RF model scores are generated                    |
| Stage 2 test   | Stage 1 outputs + Machines 4 and 5 inputs | Selected Stage 2 prediction scores are generated |
| Anomaly test   | Actual vs predicted outputs               | High-error points are flagged                    |

## 6.2 Test Procedure

The dataset was first loaded using pandas. Numeric columns were converted to numeric format, and timestamps were converted into datetime format.

The Stage 1 output columns were checked for negative values. Rows with negative values were removed because a physical output measurement should not be negative. From 14,088 rows, 35 rows were removed during this cleaning step.

Linear Regression was trained as the baseline model. Later, one-second and two-second lag features were created for process variables. Random Forest models were trained using the lag features. The models were tested using an 80% training and 20% testing split with random\_state set to 42.

Model performance was measured using Mean Absolute Error and  $R^2$  score.

## 6.3 Performance Outcome

| Stage   | Measurement  | Model             | MAE   | $R^2$ Score |
|---------|--------------|-------------------|-------|-------------|
| Stage 1 | Measurement6 | Linear Regression | 0.553 | 0.528       |
| Stage 1 | Measurement6 | Random Forest     | 0.118 | 0.867       |

| Stage   | Measurement   | Model             | MAE   | R <sup>2</sup> Score |
|---------|---------------|-------------------|-------|----------------------|
| Stage 1 | Measurement11 | Linear Regression | 0.833 | 0.724                |
| Stage 1 | Measurement11 | Random Forest     | 0.292 | 0.872                |
| Stage 1 | Measurement14 | Linear Regression | 4.459 | 0.204                |
| Stage 1 | Measurement14 | Random Forest     | 0.592 | 0.895                |
| Stage 2 | Measurement6  | Random Forest     | 0.108 | 0.458                |
| Stage 2 | Measurement11 | Random Forest     | 0.164 | 0.892                |
| Stage 2 | Measurement14 | Random Forest     | 0.512 | 0.788                |

The Random Forest model performed better than Linear Regression for all three tested Stage 1 measurements. The biggest improvement was seen for Measurement14, where the R<sup>2</sup> score increased from 0.204 with Linear Regression to 0.895 with Random Forest.

The Stage 2 test showed that some outputs can also be predicted well. Measurement11 and Measurement14 gave strong results, while Measurement6 was more difficult to predict.

For the anomaly detection prototype, Stage 1 Measurement11 was used. The anomaly threshold was set as mean prediction error plus three times the standard deviation. Out of 1,000 test points, 36 points, or 3.60%, were flagged for review.

## 7 My learnings

This internship gave practical experience in working with real industrial data. The main learning was that data cleaning is important before building any machine-learning model.

I learned how to use pandas for loading CSV files, checking data, filtering columns, calculating statistics, creating lag features, and converting timestamps. I also used NumPy for numerical operations.

I learned how to use scikit-learn for Linear Regression, Random Forest, train-test split, MAE, and  $R^2$  score. I understood that a simple model may not work for every problem. Random Forest gave better results for the selected factory output measurements because it could handle more complex patterns.

The project also helped me understand how data from machines can be used for quality prediction and anomaly detection in a real factory setting. These skills will be useful for future work in machine learning, data analysis, Industrial IoT, and predictive maintenance.

## 8 Future work scope

The following work can be done in the future:

1. Apply Random Forest and time-lag features to all 15 Stage 1 and all 15 Stage 2 output measurements.
2. Test more time delays, such as 3-second, 5-second, and 10-second lags, to identify the best delay for each output.
3. Tune Random Forest settings such as the number of trees and tree depth to improve prediction performance.
4. Test additional models such as XGBoost, Gradient Boosting, or neural network models.
5. Build a real-time dashboard that shows machine inputs, predicted output, actual output, and anomaly warnings.
6. Validate anomaly results using real fault labels or feedback from factory experts.
7. Connect the model with an industrial IoT platform for live monitoring and early quality alerts.